

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Operated and maintained professional and desktop **3D printers** for customer jobs, preparing print files and producing prototype parts, including post-processing such as cleaning and finishing.
- Conducted **3D scanning demonstrations** for enterprise customers using handheld and desktop scanners to showcase digital manufacturing solutions.
- Performed basic reverse engineering tasks, including **mesh cleanup**, alignment, and CAD recreation, supporting customers in digitizing physical components.
- Provided technical input and application advice for **pre-sales** engagements, assisting the sales team in understanding customer needs for additive manufacturing technologies.
- Responded to customer support calls, troubleshooting issues related to **3D printer operation** and print file optimization to ensure customer success.
- Authored application notes and documented **workflows and best practices** for integrating 3D printing and scanning into customer manufacturing processes.

SKILLS

3D Printing & Scanning – 3D Printer Operation, Print File Preparation, Prototype Production, Post-Processing, 3D Scanning, Reverse Engineering, Mesh Cleanup, CAD Recreation, Bambu Lab Printers

Customer Engagement – Demos, Technical Input, Application Advice, Pre-Sales Support, Support Calls, Troubleshooting, Customer Service, Workflow Documentation

Software & Tools – SolidWorks, Fusion 360, Inventor, 3D Printing Slicers, Mesh Editing Software

Manufacturing Concepts – Additive Manufacturing, Digital Workflows, Materials Science, Manufacturing Processes, Tolerances

General IT & Support – Linux, Shell Scripting, Python Scripting, SQL, Java, Spring Boot, REST APIs, AWS, Jenkins, Git, Jira, Confluence

PROJECTS

3D Printer Application Development (HCL Technologies)

- Developed and refined operational procedures for a suite of professional **3D printers**, standardizing print preparation and post-processing steps for diverse material types.
- Engineered automated Python scripts to analyze print logs and identify common failure modes, reducing troubleshooting time for production print jobs by approximately **15%**.
- Created detailed documentation and visual guides for printer maintenance and basic repair, contributing to a **20% reduction** in hardware-related downtime for the team's managed fleet.

Customer Onboarding for Digital Manufacturing Solutions (HCL Technologies)

- Collaborated with business analysts to define and document a standardized **onboarding workflow** for new clients adopting digital manufacturing tools.
- Delivered technical demonstrations of 3D scanning capabilities to prospective enterprise customers, highlighting the value proposition for reverse engineering and product development.
- Supported UAT cycles for new integration features, gathering feedback from operations teams and translating it into actionable development tickets for bug fixes and enhancements.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)